

Question	Answer	Marks
10(a)	$V_0 = \sqrt{2} \times V_{\text{r.m.s.}} = \sqrt{2} \times 9.9 \text{ (= 14 V)}$ and $\omega = 2\pi f = 2\pi \times 50 \text{ (= 314 rad s}^{-1}\text{)}$	C1
	$V = 14 \sin 314t$	A1
10(b)	enables (resonating) nuclei to be located	B1
	resonant frequency depends on magnetic field strength	B1
	Any one from: • non-uniform field is (accurately) calibrated • (non-uniform) field may be varied to enable detection in different positions • unique (magnetic) field strength/frequency at each point	B1
10(c)	$I = I_0 \exp(-\mu x)$	C1
	$I = I_0 [\exp(-\mu x)_{\text{bone}} \times \exp(-\mu x)_{\text{soft tissue}}]$	C1
	$I = I_0 [\exp(-2.9 \times 0.40) \times \exp(-0.92 \times 1.4)]$	
	$I/I_0 = 0.0865$	C1
	ratio/dB = $10 \lg 0.0865$ $= -11 \text{ dB}$	A1

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11(c)	discrete amount/quantum/packet of energy	M1